



USB 3.0

Buckle Up for Super Speed

What does the next version of USB have for us?

When I started my career a decade ago, my development PC used to have 32 MB RAM and 500 MB of hard disk space. The current generation of users can't even think of working under such configurations (or should I call it 'constraints'?).

Over the years, memory has gotten cheaper and bigger (with sizes in gigabytes now), and with different use-case models. Some of the most prominent use-case models are Flash drives that replaced floppy drives, gadgets with large memory to support media files, or camcorders with more recording time. As a user, you might want to move around large files between your PC and these devices.

With USB becoming the leading interface for PC and peripheral connectivity, speed is the next bottleneck that needs to be eliminated. As discussed, memory is getting cheaper and larger, and users are increasingly using memory devices to copy gigabyte-sized files. This would take frustratingly long hours with the existing interfaces. Also, the time taken to copy data will decrease the battery performance.

To cater to such needs, the latest USB 3.0 solution comes with a maximum speed of gigabytes in seconds, with low power consumption. Linux, for its part as a desktop

operating system, will also need to support USB 3.0 in the near future.

Let us now explore USB 3.0 from an end user's point of view, in terms of features and what has "proven" itself in Linux. For more technical insights, refer to the USB 3.0 specifications at www.usb.org.

Super speed USB 3.0

The USB 2.0 specification released in 2000 introduced high-speed devices. With the introduction of USB 3.0 in 2008, a new 'super speed' connection is being introduced. It is backward compatible with USB 2.0 and supports data rates of 5 Gbps (60 MBps). The new specification introduces new connectors (Figures 1 and 2).

Some of the salient features of the new USB specification are:

- Speed: 10x performance when compared to USB 2.0
- Power efficiency: Both of the device and host
- Backward compatible: Class drivers continue to work; legacy devices can be plugged in.

Linux support

The promoters of USB 3.0 have demonstrated USB 3.0 on Linux. They have modified the USB core and also included